WORKED SOLUTION

STEP I 1992, Question 3

Pure • Odd and Even Functions

STEP I 1992, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q3

The key idea

Every integral here is over the symmetric interval $[-\pi, \pi]$, so the first thing to ask of each integrand is whether it is even or odd. An odd function integrates to zero over a symmetric interval, while an even one gives twice the integral over $[0, \pi]$, where the modulus signs disappear. Spotting the symmetry turns all four parts into at most one short calculation each.

Solution

We use two standard facts. If $f(-x) = f(x)$ (even) then

$$\int_{-a}^{a} f(x)\,\mathrm{d}x = 2\int_{0}^{a} f(x)\,\mathrm{d}x,$$

and if $f(-x) = -f(x)$ (odd) then $\int_{-a}^{a} f(x)\,\mathrm{d}x = 0$. Both follow by substituting $x = -u$ in the integral over $[-a, 0]$, which turns it into $\int_{0}^{a} f(-u)\,\mathrm{d}u$.

We also need $\int_{0}^{\pi} \sin x\,\mathrm{d}x = [-\cos x]_{0}^{\pi} = 1 - (-1) = 2$.

Part (i)

Since $|\sin(-x)| = |-\sin x| = |\sin x|$, the integrand is even. On $[0, \pi]$ we have $\sin x \geq 0$, so $|\sin x| = \sin x$ there. Hence

$$\int_{-\pi}^{\pi} |\sin x|\,\mathrm{d}x = 2\int_{0}^{\pi} \sin x\,\mathrm{d}x = 2 \times 2 = 4.$$

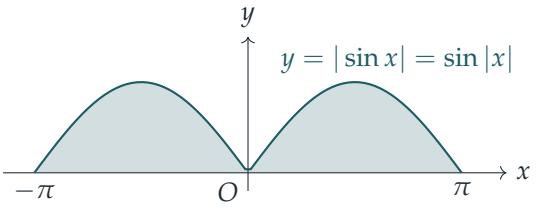
Part (ii)

Since $|-x| = |x|$, the function $\sin|x|$ is even. On $[0, \pi]$, $|x| = x$, so

$$\int_{-\pi}^{\pi} \sin|x|\,\mathrm{d}x = 2\int_{0}^{\pi} \sin x\,\mathrm{d}x = 4.$$

Ask yourself Is it a coincidence that (i) and (ii) give the same answer?

No: on this interval the two integrands are the same function. For $0 \leq x \leq \pi$ both equal $\sin x$. For $-\pi \leq x \leq 0$ we have $\sin x \leq 0$, so $|\sin x| = -\sin x$, while $\sin|x| = \sin(-x) = -\sin x$ as well. The two curves only part company outside $[-\pi, \pi]$: for example, at $x = 3\pi/2$, $|\sin x| = 1$ but $\sin|x| = -1$.



The shaded region consists of two arches of the sine curve, each of area 2.

Part (iii)

Here $(-x)\sin(-x) = (-x)(-\sin x) = x\sin x$, so the integrand is even (a product of two odd functions). Therefore

$$\int_{-\pi}^{\pi} x\sin x\,\mathrm{d}x = 2\int_{0}^{\pi} x\sin x\,\mathrm{d}x.$$

Integrating by parts with $u = x$ and $\mathrm{d}v = \sin x\,\mathrm{d}x$ (so $v = -\cos x$):

$$\begin{aligned}\int_{0}^{\pi} x\sin x\,\mathrm{d}x &= [-x\cos x]_0^{\pi} + \int_{0}^{\pi}\cos x\,\mathrm{d}x \\ &= (-\pi\cos\pi - 0) + [\sin x]_0^{\pi} \\ &= \pi + 0 = \pi.\end{aligned}$$

Hence

$$\int_{-\pi}^{\pi} x\sin x\,\mathrm{d}x = 2\pi.$$

Part (iv)

Now $(-x)^{10}\sin(-x) = x^{10}(-\sin x) = -x^{10}\sin x$, so the integrand is odd (an even function times an odd one). The integral over the symmetric interval is therefore

$$\int_{-\pi}^{\pi} x^{10}\sin x\,\mathrm{d}x = 0.$$

There is no need to integrate by parts ten times! The contributions from $[-\pi, 0]$ and $[0, \pi]$ are equal in size and opposite in sign, and they cancel exactly.

Checks

- Area sense: in (i) the integrand is non-negative and at most 1, so the answer must lie between 0 and $2\pi \approx 6.28$; 4 does.
- In (iii), $x \sin x \geq 0$ on $[-\pi, \pi]$, so the answer must be positive; 2π is. Differentiating $\sin x - x\cos x$ gives $x \sin x$, which confirms the antiderivative.
- Simpson's rule with 20 000 strips gives 4.0000000, 4.0000000, 6.2831853 and 0.0000000 for the four integrals. In (iv) the zero is not just a small number: the integral of $|x^{10} \sin x|$ over the same interval is about 13 275, so the cancellation is exact and complete.

Common slips

- Writing $\int_{-\pi}^{\pi} |\sin x| \, \mathrm{d}x = \big[-\cos x\big]_{-\pi}^{\pi} = 0$, ignoring the modulus. An integrand that is never negative cannot have integral 0 unless it is zero almost everywhere.
- Assuming $\sin|x|$ is odd because $\sin x$ is. Composing with $|x|$ always produces an even function.
- Sign errors in the integration by parts in (iii): remember $\cos \pi = -1$, so $-\pi \cos \pi = +\pi$.

Beyond STEP

Splitting a function into even and odd parts, $f(x) = \frac{1}{2}\big(f(x) + f(-x)\big) + \frac{1}{2}\big(f(x) - f(-x)\big)$, is the starting point of Fourier series. On $[-\pi, \pi]$, the even part is expanded in cosines and the odd part in sines, and integrals like (iii) and (iv) are exactly the ones used to calculate the coefficients. For example, the coefficient of $\sin x$ in the Fourier series of x comes from the integral in (iii). Boyce and DiPrima, *Elementary Differential Equations*, has a readable introduction to Fourier series and their use in solving differential equations.

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